SOAL A DATABASE

1. Buatlah tabel "users" yang memiliki struktur/model sebagai berikut. Tabel "users" merupakan kumpulan data peserta didik di sebuah sekolah.

| | # | Name | Type | Collation | Attributes | Null | Default | Comments | Extra | Action |
|---|---|---|---|---|---|---|---|---|---|---|
| ☐ | 1 | **id** | int(11) | | | No | *None* | | AUTO_INCREMENT | Change Drop More |
| ☐ | 2 | **username** | varchar(50) | utf8mb4_general_ci | | No | *None* | | | Change Drop More |
| ☐ | 3 | **email** | varchar(50) | utf8mb4_general_ci | | No | *None* | | | Change Drop More |
| ☐ | 4 | **password** | varchar(50) | utf8mb4_general_ci | | No | *None* | | | Change Drop More |
| ☐ | 5 | **created_at** | timestamp | | | Yes | current_timestamp() | | | Change Drop More |
| ☐ | 6 | **updated_at** | timestamp | | | Yes | current_timestamp() | | ON UPDATE CURRENT_TIMESTAMP() | Change Drop More |

Masukkan beberapa data berikut ke dalam tabel "users". Hasil yang diharapkan adalah:

| ←T→ | | | | id | username | email | password | created_at | updated_at |
|---|---|---|---|---|---|---|---|---|---|
| ☐ | Edit | Copy | Delete | 1 | andi | andi@andi.com | 12345 | 2025-02-12 08:45:23 | 2025-02-12 08:45:23 |
| ☐ | Edit | Copy | Delete | 2 | budi | budi@budi.com | 234234 | 2025-02-12 08:46:56 | 2025-02-12 08:46:56 |
| ☐ | Edit | Copy | Delete | 3 | caca | caca@caca.com | 2346712 | 2025-02-12 08:47:16 | 2025-02-12 08:47:16 |
| ☐ | Edit | Copy | Delete | 4 | deni | deani@deni.com | 9887363 | 2025-02-12 08:47:43 | 2025-02-12 08:47:43 |
| ☐ | Edit | Copy | Delete | 5 | euis | euis@euis.com | 877363 | 2025-02-12 08:48:05 | 2025-02-12 08:48:05 |
| ☐ | Edit | Copy | Delete | 6 | fafa | fafa@fafa.com | 344243 | 2025-02-12 08:48:22 | 2025-02-12 08:48:22 |

2. Buatlah tabel "courses" yang memiliki struktur/model sebagai berikut. Tabel "courses" merupakan kumpulan data mata kuliah yang diajarkan di sebuah sekolah.

| | # | Name | Type | Collation | Attributes | Null | Default | Comments | Extra | Action |
|---|---|---|---|---|---|---|---|---|---|---|
| ☐ | 1 | **id** | int(11) | | | No | *None* | | AUTO_INCREMENT | Change Drop More |
| ☐ | 2 | **course** | varchar(50) | utf8mb4_general_ci | | No | *None* | | | Change Drop More |
| ☐ | 3 | **mentor** | varchar(50) | utf8mb4_general_ci | | No | *None* | | | Change Drop More |
| ☐ | 4 | **title** | varchar(50) | utf8mb4_general_ci | | No | *None* | | | Change Drop More |

Masukkan beberapa data berikut ke dalam tabel "courses". Hasil yang diharapkan adalah:

| | id | course | mentor | title |
|---|---|---|---|---|
| Edit Copy Delete | 1 | C++ | Ari | Dr. |
| Edit Copy Delete | 2 | C# | Ari | Dr. |
| Edit Copy Delete | 3 | CSS | Cania | S.Kom |
| Edit Copy Delete | 4 | HTML | Kania | S.kom |
| Edit Copy Delete | 5 | Javascript | Kania | S.kom |
| Edit Copy Delete | 6 | Python | Barry | S.T. |
| Edit Copy Delete | 7 | Micropython | Barry | S.T. |
| Edit Copy Delete | 8 | Java | Darren | M.T |
| Edit Copy Delete | 9 | Ruby | Darren | M.T |

3. Buatlah tabel "userCourse" yang memiliki struktur/model sebagai berikut.
   Tabel "userCourse" merupakan tabel penghubung/transaksi antara tabel "user" & "courses".

| # | Name | Type | Collation | Attributes | Null | Default | Comments | Extra | Action |
|---|---|---|---|---|---|---|---|---|---|
| 1 | **id_user** | int(11) | | | Yes | *NULL* | | | Change Drop More |
| 2 | **id_course** | int(11) | | | Yes | *NULL* | | | Change Drop More |

   Masukkan beberapa data berikut ke dalam tabel "userCourse". Hasil yang diharapkan adalah:

| id_user | id_course |
|---|---|
| 1 | 1 |
| 1 | 2 |
| 1 | 3 |
| 2 | 4 |
| 2 | 5 |
| 2 | 6 |
| 3 | 7 |
| 3 | 8 |
| 3 | 9 |
| 4 | 1 |
| 4 | 2 |
| 4 | 3 |
| 5 | 4 |
| 5 | 5 |
| 5 | 6 |
| 6 | 7 |
| 6 | 8 |
| 6 | 9 |

4. Dari tabel "users", "courses" dan "userCourse", tampilkan semua daftar peserta didik beserta mata kuliah yang diikutinya, lengkap dengan nama & gelar mentornya. Hasil yang diharapkan adalah sebagai berikut:

```sql
SELECT
    u.id,
    CONCAT(UPPER(LEFT(u.username, 1)), LOWER(SUBSTRING(u.username, 2))) as username,
    c.course,
    c.mentor,
    c.title
FROM
    users u
    INNER JOIN usercourse uc ON u.id = uc.id_user
    INNER JOIN courses c ON uc.id_course = c.id
ORDER BY
    u.id,
    CASE
        WHEN c.course = 'C++' THEN 1
        WHEN c.course = 'C#' THEN 2
        WHEN c.course = 'CSS' THEN 3
        WHEN c.course = 'HTML' THEN 4
        WHEN c.course = 'Javascript' THEN 5
        WHEN c.course = 'Python' THEN 6
        WHEN c.course = 'Micropython' THEN 7
        WHEN c.course = 'Java' THEN 8
        ELSE 9
    END;
```


```
SELECT
  u.id,
  CONCAT(UPPER(LEFT(u.username, 1)), LOWER(SUBSTRING(u.username, 2))) as username,
  c.course,
  c.mentor,
  c.title
FROM
  users u
  INNER JOIN usercourse uc ON u.id = uc.id_user
  INNER JOIN courses c ON uc.id_course = c.id
ORDER BY
```

```
u.id,
CASE
    WHEN c.course = 'C++' THEN 1
    WHEN c.course = 'C#' THEN 2
    WHEN c.course = 'CSS' THEN 3
    WHEN c.course = 'HTML' THEN 4
    WHEN c.course = 'Javascript' THEN 5
    WHEN c.course = 'Python' THEN 6
    WHEN c.course = 'Micropython' THEN 7
    WHEN c.course = 'Java' THEN 8
    ELSE 9
END;
```

| id 1 | username | course | mentor | title |
|---|---|---|---|---|
| 1 | Andi | C++ | Ari | Dr. |
| 1 | Andi | C# | Ari | Dr. |
| 1 | Andi | C# | Ari | Dr. |
| 2 | Budi | CSS | Cania | S.Kom |
| 2 | Budi | HTML | Cania | S.Kom |
| 2 | Budi | Javascript | Cania | S.Kom |
| 3 | Caca | Python | Barry | S.T. |
| 3 | Caca | Micropython | Barry | S.T. |
| 3 | Caca | Java | Darren | M.T. |
| 4 | Deni | C++ | Ari | Dr. |
| 4 | Deni | C# | Ari | Dr. |
| 4 | Deni | HTML | Cania | S.Kom |
| 5 | Euis | C# | Ari | Dr. |
| 5 | Euis | CSS | Cania | S.Kom |
| 5 | Euis | Javascript | Cania | S.Kom |
| 6 | Fafa | Python | Barry | S.T. |
| 6 | Fafa | Micropython | Barry | S.T. |
| 6 | Fafa | Java | Darren | M.T. |

Dan ini hasilnya

| id ▲ 1 | username | course | mentor | title |
|---|---|---|---|---|
| 1 | Andi | C++ | Ari | Dr. |
| 1 | Andi | C# | Ari | Dr. |
| 1 | Andi | C# | Ari | Dr. |
| 2 | Budi | CSS | Cania | S.Kom |
| 2 | Budi | HTML | Cania | S.Kom |
| 2 | Budi | Javascript | Cania | S.Kom |
| 3 | Caca | Python | Barry | S.T. |
| 3 | Caca | Micropython | Barry | S.T. |
| 3 | Caca | Java | Darren | M.T. |
| 4 | Deni | C++ | Ari | Dr. |
| 4 | Deni | C# | Ari | Dr. |
| 4 | Deni | HTML | Cania | S.Kom |
| 5 | Euis | C# | Ari | Dr. |
| 5 | Euis | CSS | Cania | S.Kom |
| 5 | Euis | Javascript | Cania | S.Kom |
| 6 | Fafa | Python | Barry | S.T. |
| 6 | Fafa | Micropython | Barry | S.T. |
| 6 | Fafa | Java | Darren | M.T. |

5. Dari tabel "users", "courses" dan "userCourse", tampilkan daftar peserta didik beserta mata kuliah yang diikutinya, yang mentornya bergelar sarjana. Hasil yang diharapkan adalah sebagai berikut:

```
SELECT
    u.id,
    CONCAT(UPPER(LEFT(u.username, 1)), LOWER(SUBSTRING(u.username, 2))) as username,
    c.course,
    c.mentor,
    c.title
FROM
    users u
    INNER JOIN usercourse uc ON u.id = uc.id_user
    INNER JOIN courses c ON uc.id_course = c.id
WHERE
    c.title IN ('S.Kom', 'S.T.')
ORDER BY
    u.id,
    c.course;
```

SELECT

u.id,

CONCAT(UPPER(LEFT(u.username, 1)), LOWER(SUBSTRING(u.username, 2))) as username,

```
    c.course,
    c.mentor,
    c.title
FROM
    users u
    INNER JOIN usercourse uc ON u.id = uc.id_user
    INNER JOIN courses c ON uc.id_course = c.id
WHERE
    c.title IN ('S.Kom', 'S.T.')
ORDER BY
    u.id,
    c.course;
```

| id 1 | username | course | mentor | title |
|---|---|---|---|---|
| 2 | Budi | CSS | Cania | S.Kom |
| 2 | Budi | HTML | Cania | S.Kom |
| 2 | Budi | Javascript | Cania | S.Kom |
| 3 | Caca | Micropython | Barry | S.T. |
| 3 | Caca | Python | Barry | S.T. |
| 4 | Deni | HTML | Cania | S.Kom |
| 5 | Euis | CSS | Cania | S.Kom |
| 5 | Euis | Javascript | Cania | S.Kom |
| 6 | Fafa | Micropython | Barry | S.T. |
| 6 | Fafa | Python | Barry | S.T. |

6. Dari tabel "users", "courses" dan "userCourse", tampilkan daftar peserta didik beserta mata kuliah yang diikutinya, yang mentornya bergelar selain sarjana.Hasil yang diharapkan adalah sebagai berikut:

```sql
SELECT
    u.id,
    CONCAT(UPPER(LEFT(u.username, 1)), LOWER(SUBSTRING(u.username, 2))) as username,
    c.course,
    c.mentor,
    c.title
FROM
    users u
    INNER JOIN usercourse uc ON u.id = uc.id_user
    INNER JOIN courses c ON uc.id_course = c.id
WHERE
    c.title NOT IN ('S.Kom', 'S.T.')
ORDER BY
    u.id,
    c.course;
```


```sql
SELECT
  u.id,
  CONCAT(UPPER(LEFT(u.username, 1)), LOWER(SUBSTRING(u.username, 2))) as username,
  c.course,
  c.mentor,
  c.title
FROM
  users u
  INNER JOIN usercourse uc ON u.id = uc.id_user
  INNER JOIN courses c ON uc.id_course = c.id
WHERE
  c.title NOT IN ('S.Kom', 'S.T.')
ORDER BY
  u.id,
  c.course;
```


| id 1 | username | course | mentor | title |
|---|---|---|---|---|
| 1 | Andi | C# | Ari | Dr. |
| 1 | Andi | C# | Ari | Dr. |
| 1 | Andi | C++ | Ari | Dr. |
| 3 | Caca | Java | Darren | M.T. |
| 4 | Deni | C# | Ari | Dr. |
| 4 | Deni | C++ | Ari | Dr. |
| 5 | Euis | C# | Ari | Dr. |
| 6 | Fafa | Java | Darren | M.T. |

7. Dari tabel "users", "courses" dan "userCourse", tampilkan jumlah peserta didik untuk setiap mata kuliah. Hasil yang diharapkan adalah sebagai berikut:

```sql
SELECT
    c.course,
    c.mentor,
    c.title,
    COUNT(uc.id_user) as jumlah_peserta
FROM
    courses c
    LEFT JOIN usercourse uc ON c.id = uc.id_course
GROUP BY
    c.course,
    c.mentor,
    c.title
ORDER BY
    c.course;
```

```sql
SELECT
  c.course,
  c.mentor,
  c.title,
  COUNT(uc.id_user) as jumlah_peserta
FROM
  courses c
  LEFT JOIN usercourse uc ON c.id = uc.id_course
GROUP BY
  c.course,
  c.mentor,
  c.title
ORDER BY
  c.course;
```

| course ▲ 1 | mentor | title | jumlah_peserta |
|---|---|---|---|
| C# | Ari | Dr. | 4 |
| C++ | Ari | Dr. | 2 |
| CSS | Cania | S.Kom | 2 |
| HTML | Cania | S.Kom | 2 |
| Java | Darren | M.T. | 2 |
| Javascript | Cania | S.Kom | 2 |
| Micropython | Barry | S.T. | 2 |
| Python | Barry | S.T. | 2 |
| Ruby | Darren | M.T. | 0 |

8. Dari tabel "users", "courses" dan "userCourse", tampilkan jumlah peserta didik beserta total fee untuk setiap mentor. Total fee dihitung dengan besaran Rp 2.000.000,- per peserta didik. Hasil yang diharapkan adalah sebagai berikut:

```sql
SELECT
    c.mentor,
    COUNT(uc.id_user) as jumlah_peserta,
    COUNT(uc.id_user) * 2000000 as total_fee
FROM
    courses c
    LEFT JOIN usercourse uc ON c.id = uc.id_course
GROUP BY
    c.mentor
ORDER BY
    c.mentor;
```

```
SELECT
  c.mentor,
  COUNT(uc.id_user) as jumlah_peserta,
  COUNT(uc.id_user) * 2000000 as total_fee
FROM
  courses c
  LEFT JOIN usercourse uc ON c.id = uc.id_course
GROUP BY
  c.mentor
ORDER BY
  c.mentor;
```

| mentor ▲ 1 | jumlah_peserta | total_fee |
|---|---|---|
| Ari | 6 | 12000000 |
| Barry | 4 | 8000000 |
| Cania | 6 | 12000000 |
| Darren | 2 | 4000000 |